1 FIG-207 — Dft single point gate

Group: DFT-SINGLE-POINT-GATE **Evidence chain:** manifested **Status:** gate-status diagnostic; no accepted DFT result

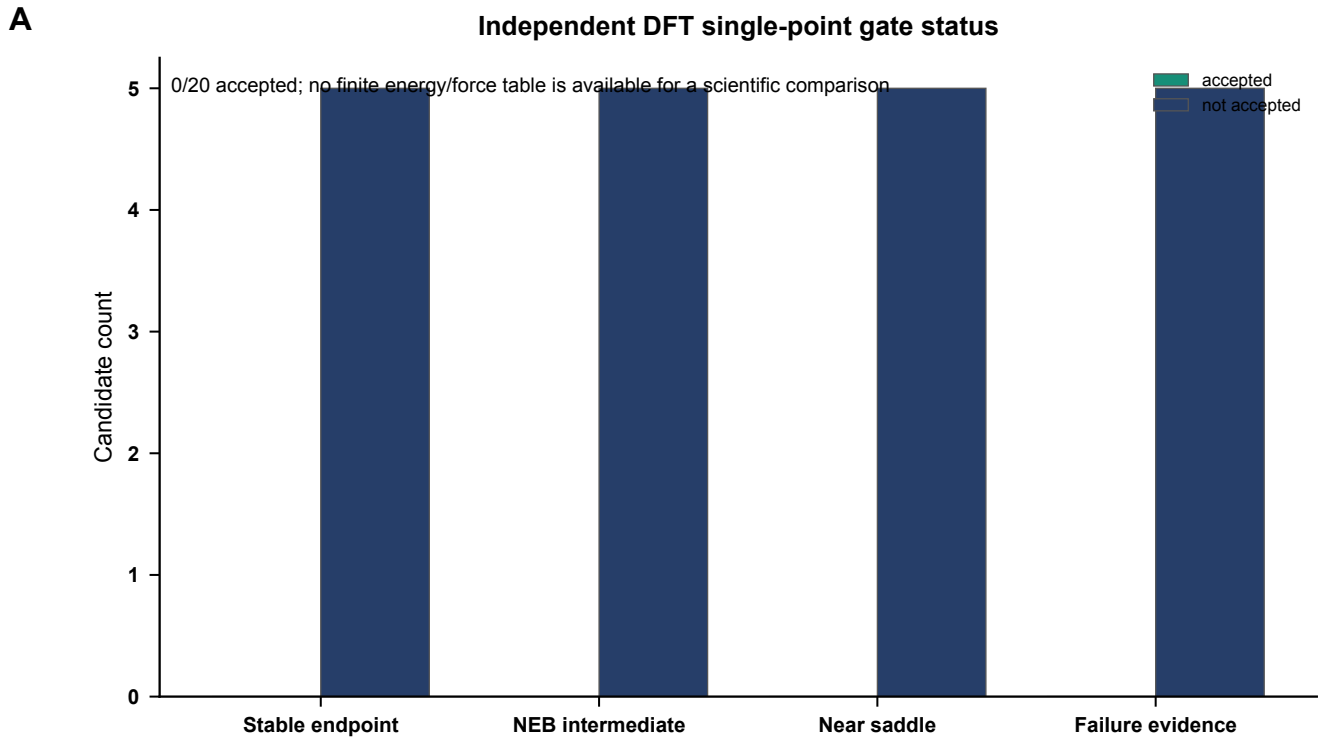


Figure 1: Candidate-level status of the independent DFT single-point wave, separated into stable endpoints, NEB intermediates, near-saddle structures, and failure-evidence frames. This page shows the Plot Atlas table-backed candidate "Dft single point gate"; its scientific status and caveats are reproduced from the package manifest.

Results text draft. All 20 audited candidates are marked not accepted: 5 stable endpoints, 5 NEB intermediates, 5 near-saddle structures, and 5 failure-evidence frames. The current table contains no accepted finite DFT energy/force values. **Discussion and boundary.** This figure is a transparent negative/gate result. It explains why MACE candidate tables cannot yet be paired with an accepted DFT target, and it prevents a missing-output campaign from being mistaken for a zero-error benchmark. **Article placement.** Proposed article location: Results/SI: DFT validation coverage and acceptance boundary. **Reviewer linkage:** overall status PARTIAL.

- **REV-A1 (PARTIAL)**: Reconcile Fig. 3d fixed-geometry DFT-path scoring with SI Fig. S3 self-consistent MLFF-NEB, including pathway identity and the DFT reference. *Completed:* The fixed-path versus self-relaxed operator distinction and the narrower model-reported-barrier wording are integrated. *Still required:* Close the restartable QE path-1–4 convergence gate, preserve the final path/source metadata, and complete the SI Fig. S3 endpoint/checkpoint/optimizer/source audit before calling the quantitative reference final.

Evidence pointers.

- Data/processed: `main/analysis/dft_single_point_validation_20261001/dft_single_point_results.csv`
- Data/processed: `main/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-SINGLE-POINT-GATE/processed_dft_gate_summary.csv`
- Plot/processing script: `main/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py`
- Caption/text source: `main/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-SINGLE-POINT-GATE/caption_results_discussion.md`

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-SINGLE-POINT-GATE/caption_results_discussion.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/dft_single_point_validation_20261001/dft_single_point_results.csv
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-SINGLE-POINT-GATE/processed_dft_gate_summary.csv
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- **Plot or processing script:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: Pending/missing QE output is not treated as a scientific zero. The MD snapshot subset also retains an explicit missing-provenance gate. This figure should not be used as a DFT energy/force plot.